

5. SRTF (preemptive)

```
# include <stdio.h>

int main() {
    int i, n, time=0, count=0;
    int at[10],bt[10],rt[10],ct[10];

    printf("Enter N: ");
    scanf("%d", &n);

    for (i=0;i<n;i++) {
        scanf("%d %d", &at[i],&bt[i]);
        rt[i]=bt[i];
    }

    while (count<n) {
        int min=999, k=-1;

        for (i=0;i<n;i++){
            if (at[i]<=time && rt[i]>0 && rt[i]<min){
                min=rt[i]; k=i;
            }
        }

        if (k==-1){
            time++;
            continue;
        }

        rt[k]--;
        time++;

        if (rt[k]==0){
            ct[k]=time;
            count++;
        }

    }

    printf("\nP\tCT\n");

    for (i=0;i<n;i++){
        printf("%d\t%d\n", i,ct[i]);
    }

    return 0;
}
```

